

GuidedProject

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Load all necessary packages

```
rm(list=ls())
```

```
# Load the packages
```

```
library(fpp3)
```

```
## -- Attaching packages ----- fpp3 1.0.3 --
```

```
## v tibble      3.3.1      v tsibble      1.2.0
## v dplyr       1.2.0      v tsibbledata 0.4.1
## v tidyr       1.3.2      v ggtime      0.2.0
## v lubridate   1.9.5      v feasts      0.5.0
## v ggplot2     4.0.2      v fable       0.5.0
```

```
## -- Conflicts ----- fpp3_conflicts --
```

```
## x lubridate::date()      masks base::date()
## x dplyr::filter()       masks stats::filter()
## x tsibble::intersect()  masks base::intersect()
## x tsibble::interval()   masks lubridate::interval()
## x dplyr::lag()          masks stats::lag()
## x tsibble::setdiff()    masks base::setdiff()
## x tsibble::union()      masks base::union()
```

```
library(codetools)
```

```
library(languageserver)
```

```
library(markdown)
```

```
library(dplyr)
```

```
library(knitr)
```

```
# Set up chunk for all slides
```

```
knitr::opts_chunk$set(
```

```
  echo = TRUE,
```

```
  message = FALSE,
```

```
  warning = FALSE,
```

```
  cache = TRUE,
```

```
  dev.args = list(pointsize = 11)
```

```
)
```

```
options(
```

```
  digits = 3,
```

```
  width = 75,
```

```
  ggplot2.continuous.colour = "viridis",
```

```
  ggplot2.continuous.fill = "viridis",
```

```
  ggplot2.discrete.colour = c("#D55E00", "#0072B2", "#009E73", "#CC79A7", "#E69F00", "#56B4E9", "#F0E442"),
```

```
  ggplot2.discrete.fill = c("#D55E00", "#0072B2", "#009E73", "#CC79A7", "#E69F00", "#56B4E9", "#F0E442")
```

```
)
```

```

##-----##
## Project: Monthly Airbnb Demand Forecasting - Fortworth
## Instructor: Dr. Jabir Rahman
##-----##

library(fpp3)
library(readr)
library(dbplyr)
library(tseries)

# 1. LOAD DATA
path_to_data <- "~/Downloads/reviews 2.csv"
raw_data <- read_csv(path_to_data)

# 2. DATA PREPARATION & MONTHLY AGGREGATION
# We aggregate the 'count' from the raw data into monthly buckets
library(lubridate)

# Convert FIRST
raw_data <- raw_data %>%
  mutate(date = ymd(date))

# Now this works
last_date <- max(raw_data$date, na.rm = TRUE)

last_complete_month <- yearmonth(
  floor_date(last_date, "month") - days(1)
)

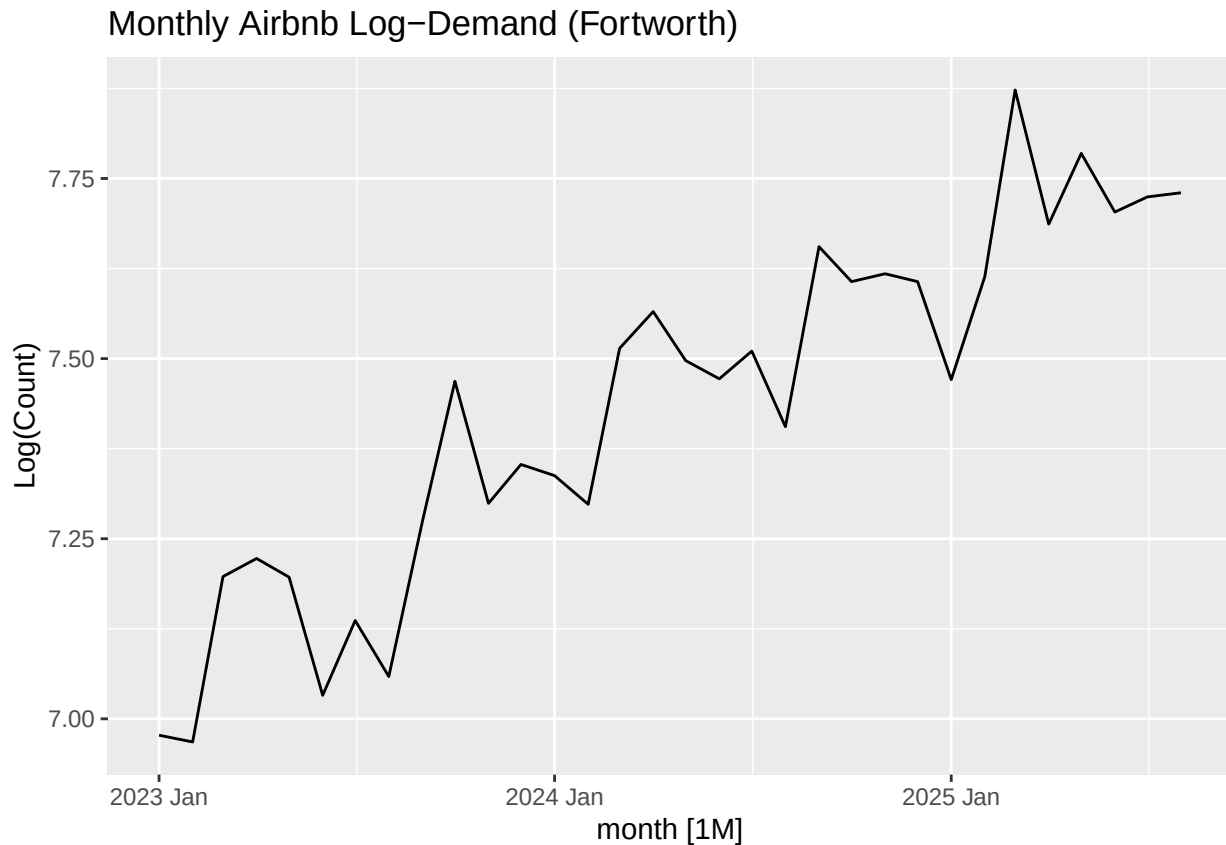
airbnb <- raw_data %>%
  mutate(month = yearmonth(date)) %>%
  count(month, name = "count") %>%
  filter(month <= last_complete_month) %>%
  as_tsibble(index = month) %>%
  mutate(log_count = log1p(count))

# 3. EXPLORATORY DATA ANALYSIS (Shortened to post-2022 for stability)
airbnb_short <- airbnb %>% filter(year(month) > 2022)
sum(is.na(airbnb_short$count))

## [1] 0

# Visualization of Trend and Seasonality
airbnb_short %>% autoplot(log_count) +
  labs(title = "Monthly Airbnb Log-Demand (Fortworth)", y = "Log(Count)")

```



```
# Stationary check (ADF test)
# Part 1 Rubric: Statistical Stationarity Check
# Using nsdiffs and ndiffs as the primary statistical evidence for stationarity
print("--- Statistical Property Check ---")
```

```
## [1] "--- Statistical Property Check ---"
```

```
stats_check <- airbnb_short %>%
  features(log_count, list(unitroot_ndiffs, unitroot_nsdiffs))
```

```
knitr::kable(stats_check, caption = "Stationarity Property Check (Required Diffs)")
```

Table 1: Stationarity Property Check (Required Diffs)

ndiffs	nsdiffs
1	1

```
library(tseries)
log_count_vec <- as.numeric(airbnb_short$log_count)
adf_result <- adf.test(log_count_vec)
print(adf_result)
```

```
##
## Augmented Dickey-Fuller Test
##
## data: log_count_vec
## Dickey-Fuller = -5, Lag order = 3, p-value = 0.01
```

```
## alternative hypothesis: stationary
kpss_result <- kpss.test(log_count_vec, null = "Level")
print(kpss_result)

##
## KPSS Test for Level Stationarity
##
## data: log_count_vec
## KPSS Level = 0.9, Truncation lag parameter = 3, p-value = 0.01

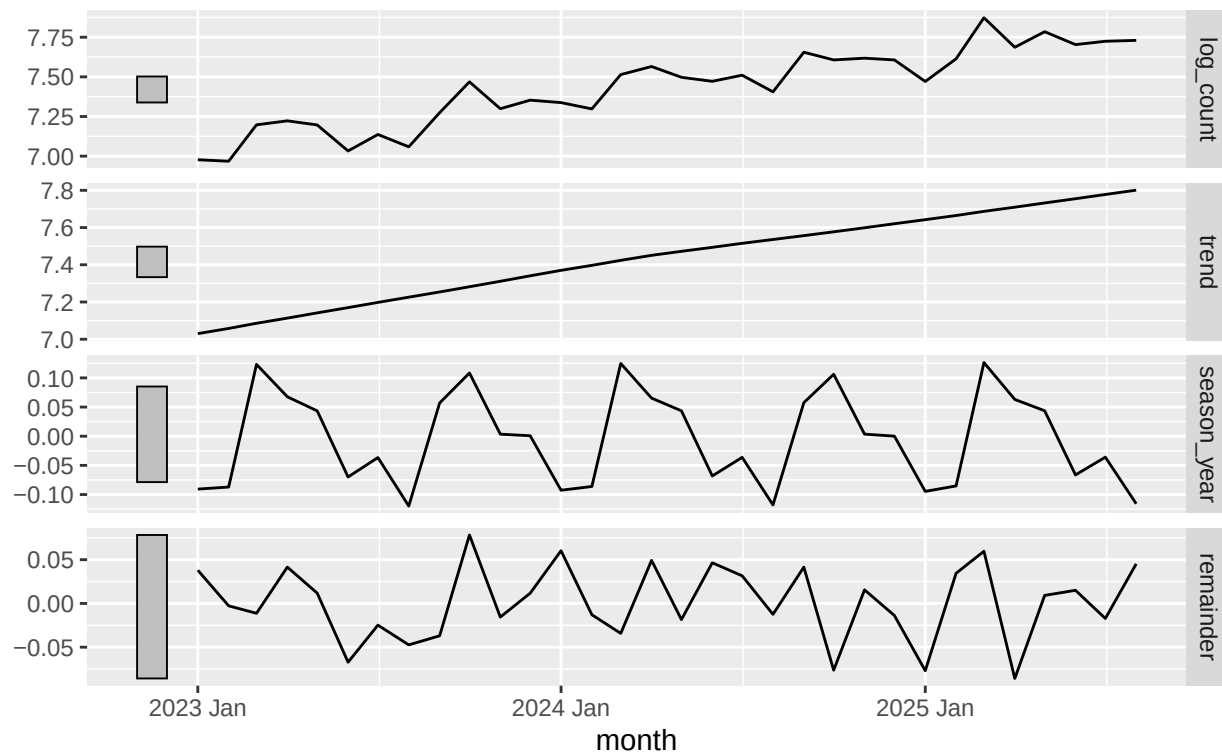
pp_result <- pp.test(log_count_vec)
print(pp_result)

##
## Phillips-Perron Unit Root Test
##
## data: log_count_vec
## Dickey-Fuller Z(alpha) = -25, Truncation lag parameter = 2,
## p-value = 0.01
## alternative hypothesis: stationary
```

```
# STL Decomposition
airbnb_short %>%
  model(stl = STL(log_count)) %>%
  components() %>%
  autoplot() + labs(title = "STL Decomposition: Trend, Seasonality, and Randomness")
```

STL Decomposition: Trend, Seasonality, and Randomness

$\log_count = \text{trend} + \text{season_year} + \text{remainder}$



```

# 4. TRAIN/TEST SPLIT
# Hold out the last 6 months for testing
h_horizon <- 6
train <- airbnb_short %>% slice(1:(n() - h_horizon))
test  <- airbnb_short %>% slice_tail(n = h_horizon)

# 5. BUILD ALL MODELS (Consolidated)
# This includes Baseline, Exponential Smoothing, ARIMA, and others
fit_all <- train %>%
  model(
    # Part 2: Baseline
    Mean = MEAN(log_count),
    `Naïve` = NAIVE(log_count),
    `Seasonal Naïve` = SNAIVE(log_count),
    Drift = RW(log_count ~ drift()),
    ETS = ETS(log_count),
    Auto_ARIMA = ARIMA(log_count),
    Manual_SARIMA = ARIMA(log_count ~ pdq(1,0,0) + PDQ(0,0,0)),
    NNETAR = NNETAR(log_count),
    Regression = TSLM(log_count ~ trend() + season()),
  )
# LJUNG-BOX test
lb_results <- fit_all %>%
  augment() %>%
  features(.innov, ljung_box, lag = 12) %>%
  rename(LB_pvalue = lb_pvalue)
# 6. FORECASTING
fc_all <- fit_all %>% forecast(h = h_horizon)

# 7. MODEL COMPARISON
accuracy_table <- fc_all %>%
  accuracy(airbnb_short) %>%
  select(.model, RMSE, MAE, MAPE, MASE) %>%
  left_join(lb_results, by = ".model") %>%
  arrange(RMSE)

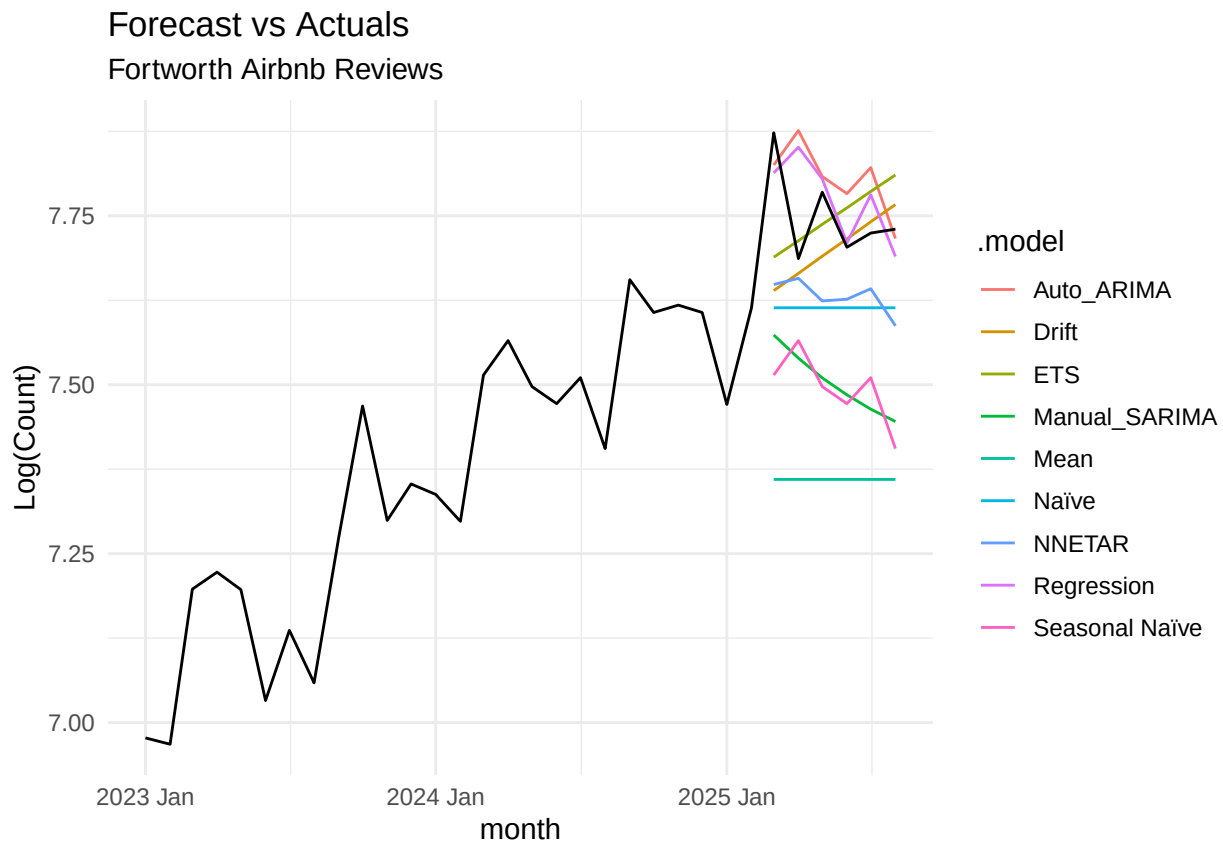
kable(
  accuracy_table,
  caption = "Table 1: Forecast Model Accuracy Comparison"
)

```

Table 2: Table 1: Forecast Model Accuracy Comparison

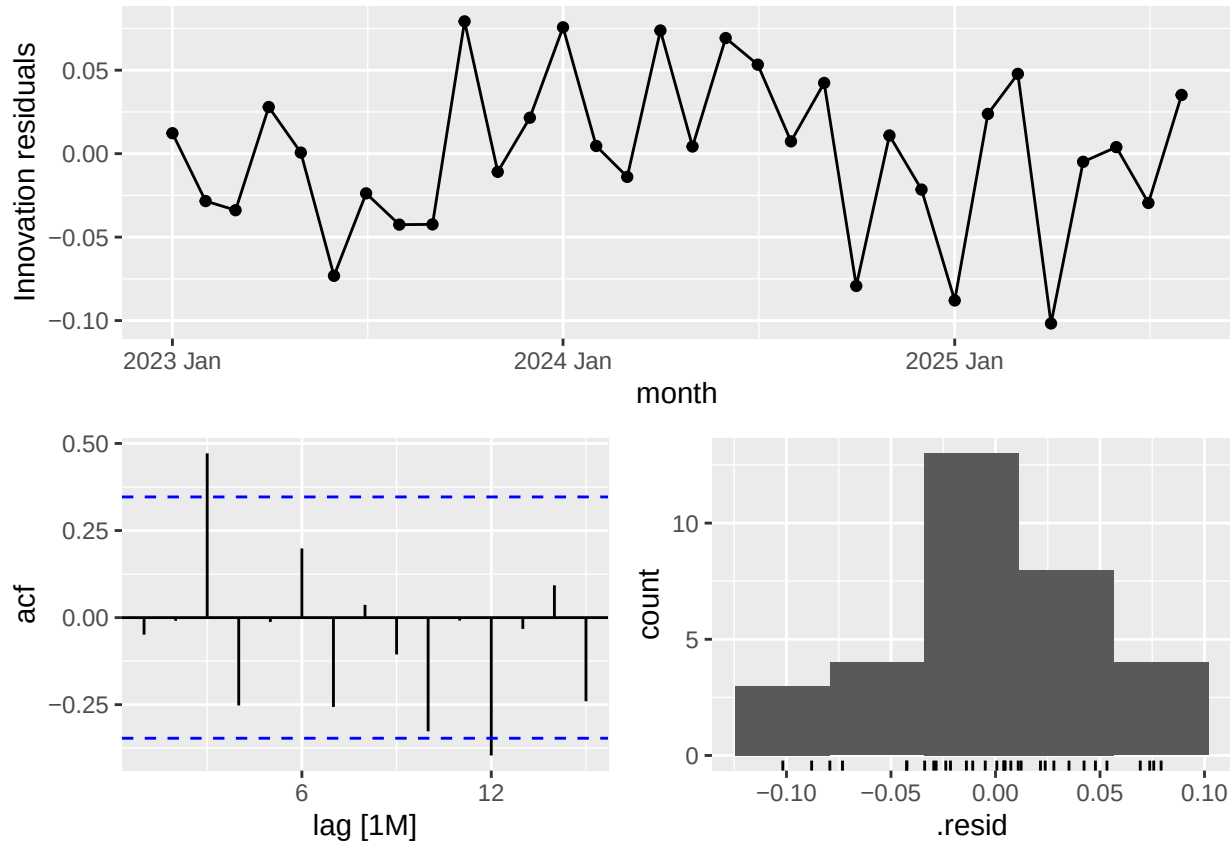
.model	RMSE	MAE	MAPE	MASE	lb_stat	LB_pvalue
Regression	0.077	0.058	0.749	0.186	24.1	0.020
ETS	0.092	0.076	0.981	0.246	37.6	0.000
Auto_ARIMA	0.095	0.075	0.972	0.241	12.3	0.423
Drift	0.105	0.069	0.885	0.223	24.2	0.019
NNETAR	0.136	0.120	1.536	0.384	7.9	0.793
Naïve	0.150	0.137	1.756	0.439	24.2	0.019
Manual_SARIMA	0.253	0.248	3.190	0.796	17.0	0.151
Seasonal Naïve	0.268	0.256	3.301	0.825	10.3	0.592
Mean	0.396	0.391	5.033	1.256	57.0	0.000

```
# 8. FINAL VISUALIZATION
fc_all %>%
  autoplot(airbnb_short, level = NULL) +
  labs(
    y = "Log(Count)",
    title = "Forecast vs Actuals",
    subtitle = "Fortworth Airbnb Reviews"
  ) +
  theme_minimal()
```



```
# FINAL MODEL
final_production_fit <- airbnb_short %>%
  model(Best_Model = TSLM(log_count ~ trend() + season()))

final_production_fit %>% gg_tsresiduals()
```



```
# 9. RESIDUAL DIAGNOSTIC TEST (Ljung-Box Test)
ljung_box_test <- final_production_fit %>%
  augment() %>%
  features(.innov, ljung_box, lag = 12)

kable(ljung_box_test, caption = "Ljung-Box Test for Residual Independence")
```

Table 3: Ljung-Box Test for Residual Independence

.model	lb_stat	lb_pvalue
Best_Model	29.8	0.003

```
final_production_fit %>%
  forecast(h = 6) %>%
  autoplot(airbnb_short) +
  labs(title = "Final 6-Month Forecast",
       subtitle = "Optimized via Regression (TSLM)",
       y = "Log(Count)") +
  theme_minimal()
```

